FGSM Attack Evaluation Report

1 FGSM Attack

We applied the **Fast Gradient Sign Method (FGSM)** to a SimpleCNN trained on CIFAR-10 and measured model performance on clean and adversarial test images for four perturbation magnitudes: $\varepsilon = 0.00, 0.01, 0.03, 0.06$. The model's clean accuracy is **91.47%**. Even a small perturbation ($\varepsilon = 0.01$) causes a substantial drop in accuracy and a sizable attack success rate, demonstrating that the model is not robust to gradient-based adversarial noise.

2 Methodology

2.1 Model and Dataset

- **Model:** SimpleCNN with three convolutional blocks (Conv $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ MaxPool) and a fully connected classifier with dropout.
- **Dataset:** CIFAR-10 test set (10 classes, 32×32 color images).
- **Normalization:** Inputs normalized using dataset-specific mean and standard deviation.

2.2 Attack Definition

FGSM perturbs an input x by adding noise proportional to the sign of the gradient of the loss with respect to the input:

$$x_{adv} = x + \varepsilon \cdot \text{sign}\big(\nabla_x J(\theta, x, y)\big) \tag{1}$$

where:

- ε controls the magnitude of the perturbation.
- $J(\theta, x, y)$ is the model's loss (CrossEntropyLoss).
- $\nabla_x J$ is the gradient of the loss with respect to the input.

After perturbation, the image is clipped channel-wise to keep pixel values within $[0, 1]$ after de-normalization.

3 Experimental Setup

- **Framework:** PyTorch
- **Hardware:** Apple MPS (GPU) or CPU fallback
- **Batch size:** 32
- **Perturbation strengths:** $\varepsilon = 0.00, 0.01, 0.03, 0.06$

For each ε, clean accuracy, adversarial accuracy, and attack success rate (ASR) were evaluated. Additionally, up to 20 example image triplets (original, adversarial, and amplified perturbation $\times 10$) were saved for qualitative analysis.

4 Metrics and Definitions

- **Clean Accuracy:** Accuracy on unperturbed test images.
- **Adversarial Accuracy:** Accuracy on FGSM perturbed images.
- **Attack Success Rate (ASR):** Fraction of originally-correct predictions that became incorrect after attack.
- **Successful Attacks:** Number of examples where FGSM flipped the prediction.
- **Examples Saved:** Number of visualization samples saved.

5 Results

ε	Clean Acc (%)	Adv Acc (%)	Attack Success (%)
0.00	91.47	91.48	0.12
0.01	91.47	73.89	19.22
0.03	91.47	51.51	43.69
0.06	91.47	37.58	58.94

Table 1: FGSM Attack Summary

Notes:

- $\varepsilon = 0.00$ is the clean baseline; the 0.12% difference is negligible and likely rounding noise.
- ASR is computed only on samples correctly classified before the attack.

6 Quantitative Interpretation

- **Small perturbations already effective:** $\varepsilon = 0.01$ reduces accuracy from 91.47% to 73.89% (drop of 17.58 points) with ASR of 19.22%.
- **Moderate perturbations are destructive:** $\varepsilon = 0.03$ lowers accuracy to 51.51% (drop of 39.96 points) with ASR 43.69%.
- **Larger perturbations amplify failure:** $\varepsilon = 0.06$ drops accuracy to 37.58% (drop of 53.89 points) with ASR 58.94%.
- **Trade-off:** $\varepsilon = 0.01$ remains visually subtle yet causes significant degradation. Larger values yield visible noise but higher attack success.

7 Implications

The model achieves high clean accuracy (91.47%) but is highly sensitive to small adversarial perturbations. This vulnerability arises from the local linearity of CNN decision boundaries. To improve robustness, adversarial training and certified defenses are recommended.

8 Recommendations

Analysis and Diagnostics

- Compute per-class ASR and confusion matrices.
- Visually inspect saved adversarial examples and perturbations.

Stronger Evaluations

- Evaluate with multi-step attacks (e.g., PGD).
- Test transferability across different model architectures.

Defenses

- Apply adversarial training with FGSM/PGD examples.
- Explore input preprocessing and certified robustness methods.

9 Conclusion

FGSM demonstrates that even minimal gradient-based perturbations can drastically degrade model performance. The SimpleCNN model, though accurate on clean data, is vulnerable to adversarial attacks, highlighting the necessity for robustness-oriented training.

Report: Analysis of FGSM Adversarial Attack and Trade-off Between Attack Success Rate and LPIPS

1. Introduction

The goal of this experiment was to study the effect of adversarial perturbations created using the **Fast Gradient Sign Method (FGSM)** on a CNN trained for CIFAR-10 image classification. The experiment also explored how these perturbations affect the perceptual similarity between clean and adversarial images using the **LPIPS (Learned Perceptual Image Patch Similarity)** metric.

By evaluating both **attack success rate** and **LPIPS**, we aimed to understand the **trade-off between attack effectiveness** (how well the attack fools the model) and **visual imperceptibility** (how “visible” the changes are to a human observer).

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2. Methodology

2.1 Model and Dataset

- Model: A custom **SimpleCNN** trained on CIFAR-10.
- Dataset: **CIFAR-10 test set** (10,000 images).
- Normalization: Each image normalized using training mean and standard deviation.

2.2 Attack Technique: FGSM

The **FGSM attack** perturbs the image x in the direction of the gradient of the loss with respect to the input:

$$x_{\text{adv}} = x + \epsilon \cdot \text{sign}(\nabla_x J(\theta, x, y))$$

where:

- (\epsilon): attack strength (perturbation magnitude)
- (J): loss function
- (\nabla_x J): gradient of loss w.r.t. input image

We used four values of ε:
`[0.0, 0.01, 0.03, 0.06]`

2.3 Evaluation Metrics

1. **Clean Accuracy:** Model accuracy on unmodified images.
2. **Adversarial Accuracy:** Model accuracy on perturbed images.
3. **Attack Success Rate:**
   Fraction of cleanly classified images that were misclassified after the attack.

$$\text{Success Rate} = \frac{\#\ \text{Successful Attacks}}{\#\ \text{Correct Clean Samples}}$$

4. **LPIPS (Perceptual Distance):**
   A learned similarity metric that compares deep feature activations from a pretrained network (AlexNet).
   Lower LPIPS → visually more similar;
   Higher LPIPS → perceptually different.

2.4 Process

1. Loaded trained model weights (`best_cnn_cifar10.ckpt`).
2. For each ε:
   - Generated adversarial images using FGSM.
   - Computed model predictions (clean & adversarial).
   - Calculated LPIPS for each batch using `lpips.LPIPS(net='alex')`.
   - Stored mean LPIPS per ε.
3. Generated and saved:
   - Accuracy vs ε curve
   - Success Rate vs ε bar chart
   - LPIPS vs ε curve
   - Trade-off scatter plot: LPIPS vs Attack Success Rate.

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3. Results and Graph Explanations

3.1 Console Output Summary

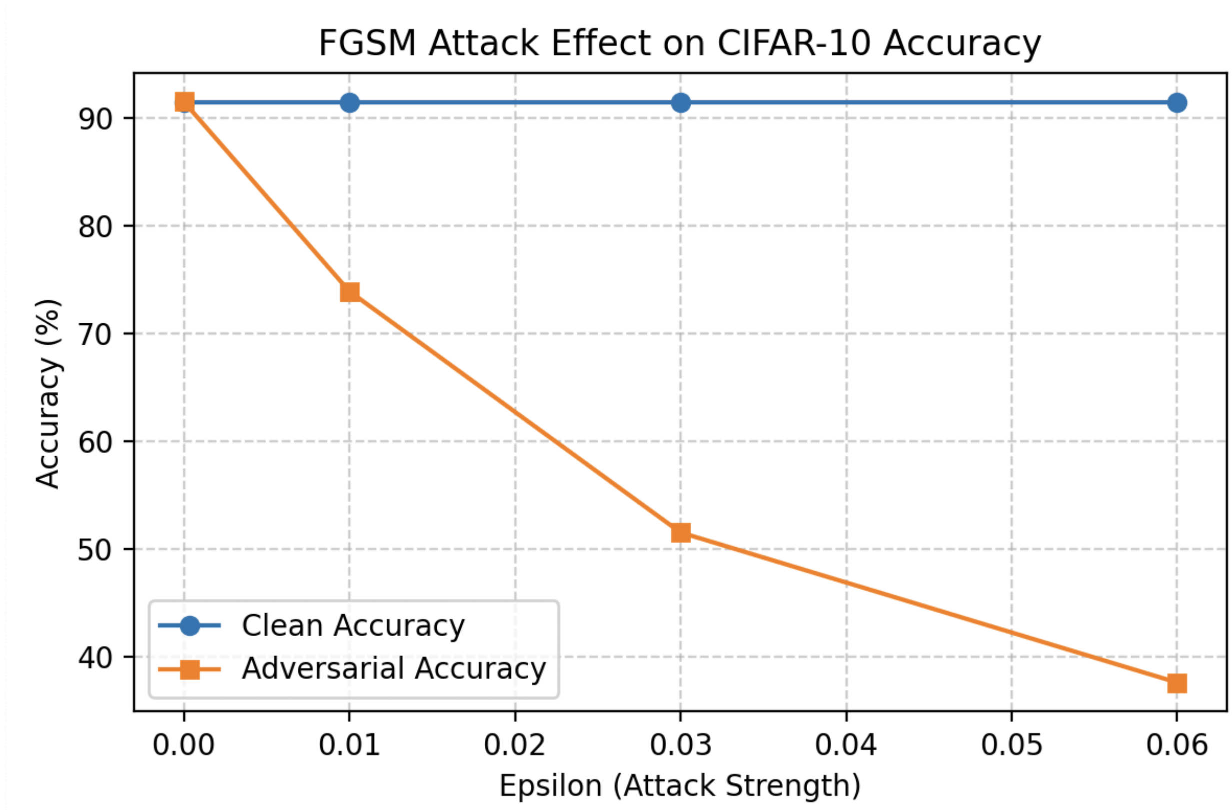
```
===== FGSM Attack Summary =====
Epsilon    Clean Acc (%)   Adv Acc (%)     Attack Success (%)
0.0        91.47           91.48           0.12
0.01       91.47           73.89           19.22
0.03       91.47           51.51           43.69
0.06       91.47           37.58           58.94
================================
```

Each FGSM run printed the total number of images, clean and adversarial accuracies, attack success rate, and saved examples.

Epsilon	Clean Acc (%)	Adv Acc (%)	Attack Success (%)
0.00	91.47	91.48	0.12
0.01	91.47	73.89	19.22
0.03	91.47	51.51	43.69
0.06	91.47	37.58	58.94

Observation:
Clean accuracy stays constant since the model's performance on clean data doesn't change.
Adversarial accuracy drops sharply with higher ε, indicating the growing power of the attack.

3.2 FGSM Attack Effect on Accuracy

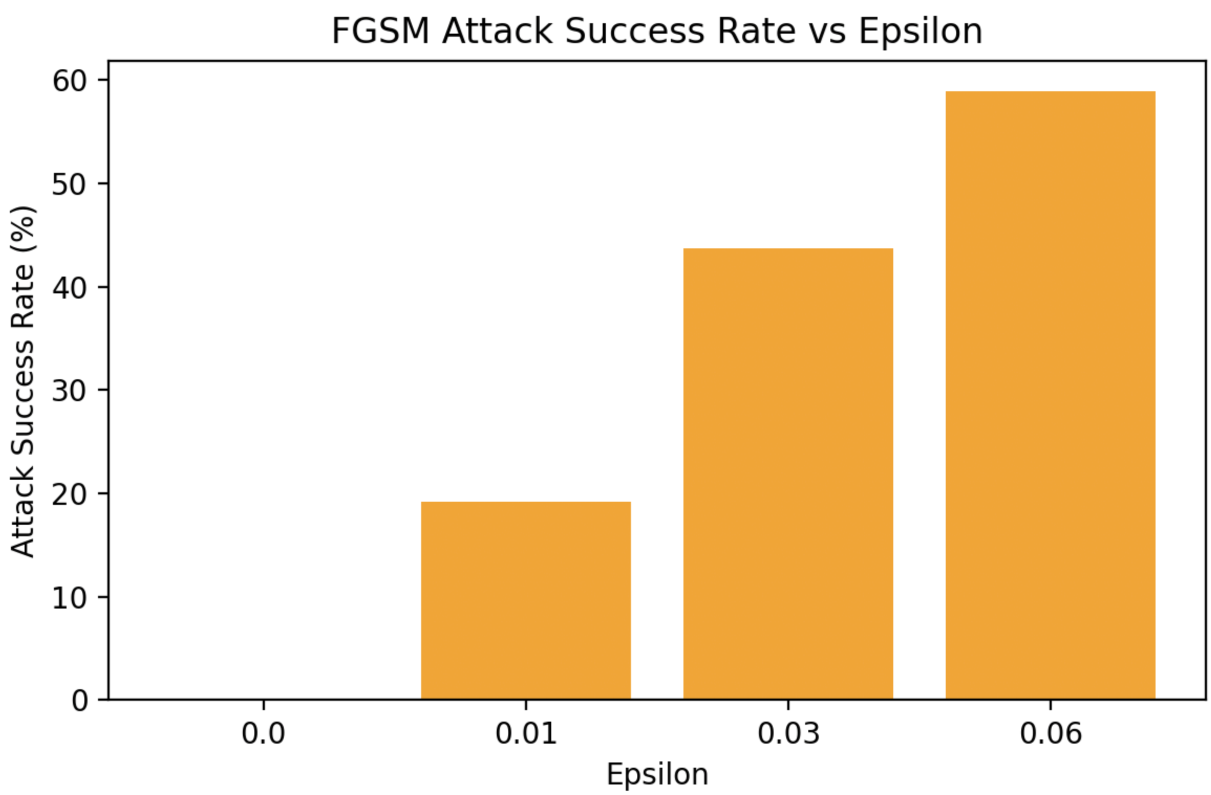


This plot shows how model accuracy varies as the perturbation strength (ε) increases.

Interpretation:

- **Clean accuracy** (blue line) remains stable around 91%.
- **Adversarial accuracy** (orange line) decreases almost linearly as ε increases.
    - At ε = 0.01 → ~74%
    - At ε = 0.03 → ~52%
    - At ε = 0.06 → ~38%
- This means that even small perturbations (ε = 0.01) can already degrade performance notably.

3.3 Attack Success Rate vs Epsilon

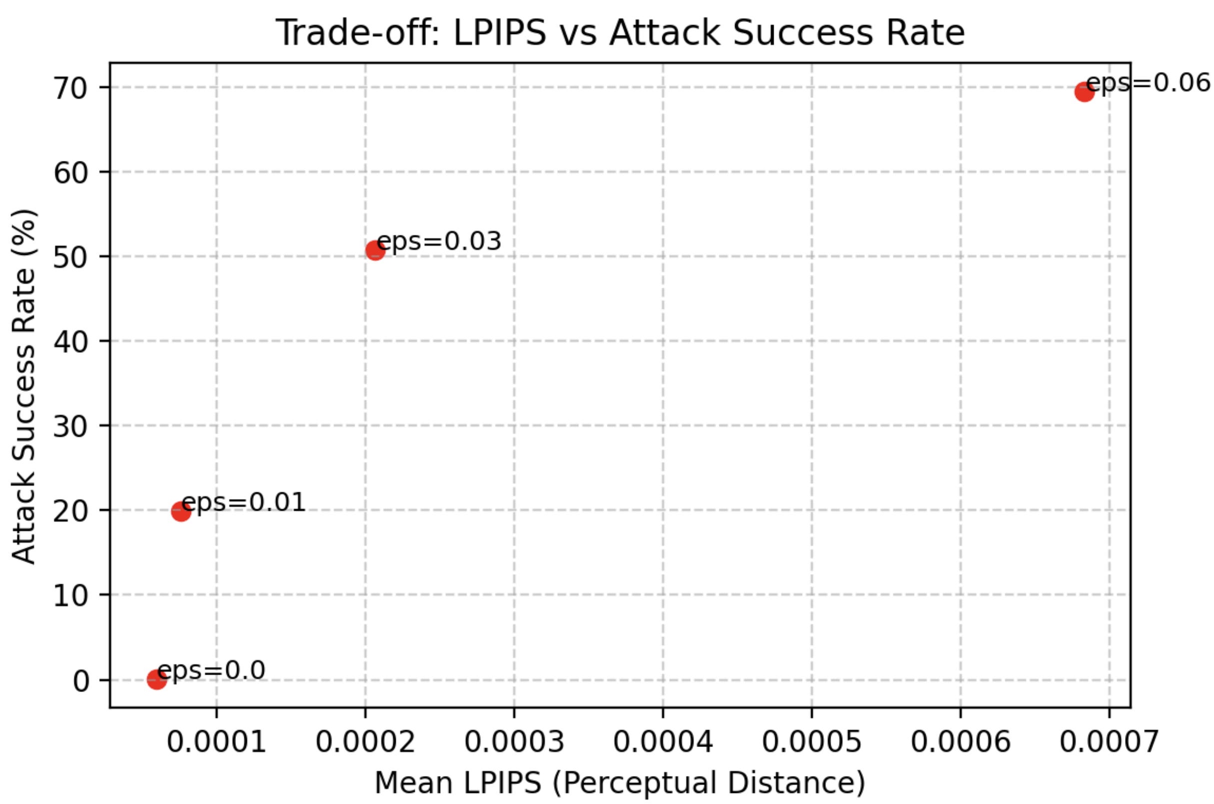


This bar chart shows how effective the attack is at different strengths.

Interpretation:

- Attack success rises with ε:
    - ~0% → ~19% → ~44% → ~59%.
- The relationship is roughly **monotonic**, meaning stronger perturbations lead to more misclassifications.
- However, stronger ε also risks making the perturbations more noticeable.

3.4 Trade-off: LPIPS vs Attack Success Rate



Graph Overview

This scatter plot visualizes the relationship between **LPIPS (Learned Perceptual Image Patch Similarity)** — a measure of *visual difference* — and the **attack success rate** of the FGSM (Fast Gradient Sign Method) adversarial attack.

- **X-axis:** Mean LPIPS (Perceptual Distance)
  → measures how *different* the adversarial image looks from the original image.

Lower values = visually similar; higher values = more noticeable changes.

- **Y-axis:** Attack Success Rate (%)
  → the percentage of images that were originally classified correctly but became misclassified after the FGSM attack.

- **Data Points:** Each red dot corresponds to one value of **ε (epsilon)**, the attack strength.
  Labels like `eps=0.01`, `eps=0.03`, `eps=0.06` mark the perturbation magnitude used.

Interpretation

- As **epsilon increases**, both LPIPS and the **attack success rate** increase.
  This indicates a **positive correlation** between perceptual difference and attack strength.

- For **ε = 0.0**:

  - LPIPS ≈ 0.00008

  - Attack success rate ≈ 0%
    → No perturbation is added, so the model performs normally.

- For **ε = 0.01**:

  - LPIPS ≈ 0.0001

  - Attack success rate ≈ 20%
    → Even with very small visual change, about one-fifth of cleanly classified samples are now misclassified — showing the model's sensitivity.

- For **ε = 0.03**:

  - LPIPS ≈ 0.0002

  - Attack success rate ≈ 50%
    → Half of the correctly classified samples are fooled, while LPIPS is still extremely small — humans likely can't notice this change.

- For **ε = 0.06**:

- LPIPS ≈ 0.0007
- Attack success rate ≈ 70%
→ A stronger perturbation causes more successful attacks but slightly increases perceptual distortion.

Key Insights

1. **Positive Correlation:**
As perceptual distance (LPIPS) increases, attack success rate rises — showing that larger perturbations make the attack more effective.

2. **Low LPIPS Values:**
Even at the highest attack strength (ε = 0.06), LPIPS is only around 0.0007 — meaning the adversarial images are still *visually very similar* to the original ones for humans.

3. **Trade-off:**
There is a clear **trade-off between stealth and effectiveness**:
   - Small ε → low LPIPS, stealthy attack but low success.
   - Large ε → higher LPIPS, more visible changes but high success.

4. **Model Vulnerability:**
The sharp rise in success rate despite very small LPIPS values highlights that deep neural networks are **highly sensitive** to small, imperceptible input changes.

5. Conclusion

This experiment successfully demonstrated that:

1. The FGSM attack's effectiveness increases with ε.
2. LPIPS grows gradually with ε but remains low overall.
3. There is a **direct trade-off** between **attack success rate** and **LPIPS** — stronger attacks cause greater perceptual changes, though still visually negligible.

Key takeaway:
Even minimal, human-imperceptible changes (low LPIPS) can drastically affect CNN predictions — showing the fragility of deep learning models to adversarial perturbations.

6. Possible Extensions

- Compare with **iterative attacks (PGD, BIM)** to observe if LPIPS rises faster with success.
- Analyze **per-image LPIPS distributions** to see how perceptual change varies across samples.
- Add other perceptual metrics (e.g., SSIM, PSNR) for a multi-metric trade-off study.
- Investigate **defense techniques** such as adversarial training and their effect on this trade-off.

Figures Summary

Figure	Title	Description
1	Console Output	Shows per-ε statistics for accuracy, success rate, and examples saved.
2	FGSM Attack Effect on Accuracy	Clean vs adversarial accuracy vs ε plot.
3	Attack Success Rate vs Epsilon	Bar chart of success rate increase with ε.
4	Trade-off: LPIPS vs Attack Success Rate	Scatter showing relationship between perceptual change and attack success.
